

Invisio Test Plan Report

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1. Introduction

This test plan explains how the project titled **Orion AI: Invisio** will be evaluated through systematic testing. The main purpose of this document is to ensure that the system operates as intended and satisfies the defined functional requirements. The tests focus on key features such as image upload, AI-based editing operations, preview generation, and downloading of processed images. The results of these tests will help detect possible issues and confirm the system's reliability and stability prior to final submission.

2. Project Scope

2.1. Project title:

Invisio: AI-Powered Image Editing and Processing System

2.2. Brief description of the system:

Invisio is a web-based application that uses artificial intelligence to perform automated image editing and enhancement tasks. The system enables users to upload images and apply various AI-driven operations such as object removal, image enhancement, background editing, style transfer, and colorization. In addition to fully automated processing, the system also supports limited manual refinement to improve AI-generated results when necessary.

The application is designed to simplify complex image editing tasks for non-technical users through an intuitive interface. Users can preview the processed output and download the final image without requiring professional editing skills. All image data is handled within temporary sessions and is removed after processing, ensuring privacy and secure data handling.

2.3. Main functionalities of the system

The main functionalities provided by the system include:

- User registration and login authentication
- Secure user session management
- Uploading image files via a web interface
- Validation of uploaded images (format and size checks)
- AI-based image processing and editing operations
- Automatic object detection and removal
- Image enhancement (brightness, contrast, and color adjustment)
- Artistic style transfer using neural models
- Facial beautification in portrait images
- Automatic background removal, replacement, or blurring
- Colorization of black-and-white images
- Preview of processed results before confirmation
- Downloading the final edited image
- Manual refinement of AI-generated masks using brush-based tools

- Support for a hybrid editing workflow combining automation with user refinement
- Session-based processing with temporary data storage for privacy

These functionalities represent the core capabilities of the system and will be evaluated using different testing methods during the testing phase.

3. Features to Be Tested

Requirement ID	Requirement Description
FR-01	The system shall allow users to upload image files through a web-based interface.
FR-02	The system shall automatically detect and remove unwanted objects from images using AI-based inpainting techniques.
FR-03	The system shall enhance image properties such as brightness, contrast, and color using AI-driven models.
FR-04	The system shall apply artistic styles to images through neural style transfer methods.
FR-05	The system shall enhance facial features in portrait images using AI-based beautification algorithms.
FR-06	The system shall automatically remove, replace, or blur image backgrounds using segmentation techniques.
FR-07	The system shall convert black-and-white images into colorized versions using AI models.
FR-08	The system shall display a preview of processed images before final confirmation.

FR-09	The system shall allow users to download the edited images in a supported format.
FR-10	The system shall process image data within a temporary session and delete it after processing is completed.
FR-11	The system shall allow users to manually refine AI-generated masks using brush-based tools.
FR-12	The system shall support a hybrid workflow combining automated AI processing with optional manual refinement.
FR-13	The system shall detect and reject NSFW or restricted content before processing.

4. Testing Methodology

The following testing methods are used in this project to ensure that the system works correctly, reliably, and efficiently. Different testing strategies are applied at various stages of development in order to evaluate individual components as well as the complete system.

Automated backend tests are implemented using a Python testing framework that simulates API requests and validates system responses. Test image files are generated programmatically to ensure consistent and reproducible testing conditions.

4.1. Unit Testing

Unit testing is used to verify the functionality of individual modules and functions in the system. Each component is tested independently to ensure that it produces correct outputs for given inputs. This helps detect implementation errors at an early stage of development.

In this project, unit tests focus on isolated backend components such as:

- Image validation: Ensuring that valid image formats are accepted and unsupported formats are rejected.
- AI processing modules: Verifying that individual AI editing modules produce valid processed image outputs.
- Authentication logic: Confirming that user registration, login, and credential verification behave correctly.
- Session management: Ensuring that temporary sessions are created, retrieved, and deleted correctly.

4.2. Integration Testing

Integration testing focuses on verifying the interaction between different system modules. This stage ensures that separate modules work together smoothly as part of the complete system.

Examples of integration testing include:

- Verifying that uploaded images are correctly transferred from the upload module to the AI processing module.
- Ensuring that processed images can be retrieved through the preview and download components.
- Testing the interaction between authentication mechanisms and protected system resources.

4.3. System Testing

System testing evaluates the entire application as a complete system. In this phase, the full workflow of the application is tested from start to finish.

The typical system workflow includes:

- Uploading an image to the system
- Selecting an AI editing feature
- Processing the image using the selected AI model
- Displaying the processed result through the preview interface
- Downloading the final edited image

This testing ensures that all system components function together correctly and satisfy the defined project requirements.

4.4. Security Testing

Security testing evaluates the system's ability to handle malicious inputs and unauthorized access.

The following aspects are tested:

- Input validation: Ensuring that invalid or corrupted files are rejected by the system.
- Access control: Verifying that protected features require proper authentication.
- Request protection: Confirming that the system limits excessive requests to prevent abuse.
- Content filtering: Ensuring that restricted or inappropriate image content is detected and blocked before processing.

4.5. Performance Testing

Performance testing is conducted to evaluate the system's speed and efficiency when processing images.

The following aspects are evaluated:

- The time required to upload images of different sizes
- The time required for AI-based image editing operations
- The system's ability to handle multiple requests simultaneously
- The efficiency of session cleanup and temporary data removal

These tests help ensure that the system performs reliably under realistic workloads.

4.6. User Testing

User testing evaluates the system from the perspective of end users.

- Sample users interact with the application by performing common tasks such as:
- Uploading images and applying AI editing features
- Adjusting results using manual refinement tools
- Navigating the hybrid workflow of automated editing and manual correction
- Downloading final edited images
- User feedback is collected regarding usability, interface clarity, and overall satisfaction with the editing results.

5. Test Environment

This section describes the hardware and software environment used for testing the system. The goal is to ensure that the testing process reflects the actual conditions under which the application will operate.

5.1. Hardware

The system is tested on a standard personal computer environment capable of handling AI-based image processing tasks.

- CPU: AMD Ryzen 9 5900HX with Radeon Graphics 8-Core CPU
- RAM: 16 GB
- GPU: NVIDIA GeForce RTX 3060 6 GB Laptop GPU

5.2. Software

The system is tested using the following software technologies and development tools:

General:

- Operating System: Windows
- Version Control System: Git / GitHub

Backend:

- Programming Language: Python
- Web Framework: FastAPI 0.115
- ASGI Server: Uvicorn 0.34
- ORM: SQLAlchemy 2.0 (async)
- Data Validation: Pydantic 2.10
- Image Processing: Pillow 11.1
- Authentication: JWT (python-jose), Passlib + bcrypt
- Testing: pytest, httpx

AI / Deep Learning:

- PyTorch (optional GPU support)
- NumPy

Database & Infrastructure:

- Database: PostgreSQL 15
- Containerization: Docker / Docker Compose
- Async Driver: asyncpg

Frontend:

- UI Framework: React 18.2
- Type System: TypeScript 5.9
- Build Tool: Vite 4.5
- CSS Framework: Tailwind CSS 3.4
- State Management: Zustand 5.0
- Routing: React Router 7.13
- HTTP Client: Axios
- Animations: Framer Motion

These tools and technologies form the core development and testing environment of the system and ensure that the application can be tested and deployed consistently across different machines.

6. Test Cases

Each test case should be identified with a unique Test ID.

Test ID	Description	Input(s)	Expected Output	Result
TC-01	Verify that the system allows users to upload a valid image file.	Valid image file (JPG/PNG).	The image is successfully uploaded and displayed.	Passed
TC-02	Verify that the system rejects unsupported file formats.	Invalid file type (.txt, .pdf).	The system rejects the file and shows an error message.	Passed

TC-03	Verify AI processing workflow for a selected editing operation.	Uploaded image + selected AI editing feature.	The AI module processes the image and generates an edited result.	Pending
TC-04	Verify that processed results are displayed correctly in preview.	Processed image from AI module.	The preview screen shows the edited image correctly.	Passed
TC-05	Verify downloading of processed images.	Processed image ready for export.	Image is downloaded successfully and matches preview.	Passed
TC-06	Evaluate system performance with large image files.	High-resolution image.	Image processed within acceptable time.	Pending
TC-07	Verify session-based temporary image handling.	Uploaded image during active session.	Image deleted after session completion.	Pending
TC-08	Verify system behavior when processing is requested without image upload.	Processing request without image.	The system displays an error message.	Passed
TC-09	Verify object removal feature using AI inpainting.	Image containing removable object.	Object removed and background reconstructed.	Pending
TC-10	Verify image enhancement feature.	Low-quality image.	The system improves brightness, contrast, or color balance.	Pending
TC-11	Verify neural style transfer feature.	Uploaded image + selected artistic style.	The system generates a stylized version of the image.	Pending
TC-12	Verify facial beautification feature.	Portrait image containing face.	The system enhances facial features naturally.	Pending
TC-13	Verify background editing functionality.	Image with background.	Background removed, blurred, or replaced.	Pending

TC-14	Verify black-and-white image colorization.	Grayscale image.	The system generates colorized images.	Passed
TC-15	Verify manual mask refinement tool.	AI-generated mask + user brush input.	Mask updated according to user corrections.	Pending
TC-16	Verify hybrid AI + manual editing workflow.	Masked image by AI + refinement adjustments.	System reprocesses image using corrected mask.	Pending
TC-17	Verify NSFW or restricted content detection.	The user uploads an image containing inappropriate or restricted content.	The system detects the NSFW content, blocks further processing, and displays an appropriate warning message. The image is not processed or stored.	Passed
TC-18	Verify community sharing feature.	The user chooses to share the final image.	Image published to community gallery without metadata.	Pending
TC-19	Verify user authentication with the sign in and login screen.	The user enters personal information to sign in or login.	The user creates an unique account and reaches it with personal information.	Passed

7. Risks and Limitations

7.1. Possible technical risk

There are technical risks that may arise during the development and testing process of the system. These risks directly affect the system's performance, reliability, and user experience. Therefore, identifying these risks early in the development process and taking necessary precautions is crucial.

- One technical risk relates to the performance of artificial intelligence models. Since image processing operations require high computational capacity, running AI algorithms can lead to intensive use of system resources. These operations may take longer than expected, especially on computers with low processing power or systems without GPU support. This can increase the system's response time and negatively impact the user experience.
- Another issue among the technical risks of the system concerns the harmonious functioning of independently operating components. The system consists of multiple functional layers, such as media loading, AI-assisted visual analysis, preview presentation, and output transfer. Inconsistent

or faulty management of data transfers between these layers can have negative consequences for the entire system. In particular, disruptions in communication between components can lead to unexpected system behaviors, process interruptions, or the inability to complete certain processes.

- Processing large image files also poses another technical risk for the system. When users upload high-resolution or large images, the system's memory usage can increase. If these files are not managed properly, system performance may decrease and processing times may increase. Therefore, optimizing image sizes and carefully managing memory are important.
- Users uploading incorrect file types or misusing the system can lead to errors. Therefore, it is important that the system is designed to detect user errors and provide appropriate warnings.
- The image filters used in the system may require high processing power in some cases. Applying filters, especially to high-resolution images, can increase processing time and negatively affect system performance. This may cause users to receive results later.
- Lastly, security and privacy issues are among the important risks to consider. Since the system processes images uploaded by users, this data must be protected securely. Unauthorized storage or sharing of user data with third parties can lead to serious security problems. Therefore, careful implementation of data security, access control, and data storage policies in the system is of great importance.

7.2. Dataset or hardware limitations

- The success of AI-based image processing systems largely depends on the quality of the datasets used and the available hardware resources. Therefore, the system's performance can be affected by certain data and hardware limitations.
- One of the most significant hardware limitations is the lack of GPU access. AI models generally run much faster on graphics processing units (GPUs). If the system is run in an environment that uses only the CPU, image processing operations may take longer and system performance may decrease. This can be especially noticeable when processing a large number of images.
- Another limitation is related to the generalization ability of AI models. Since AI models are usually trained on specific datasets, they may not always produce results with the same accuracy on images with different characteristics. For example, very low-resolution images, excessively dark or bright images, or images with complex backgrounds can negatively affect the model's performance. Therefore, the system may not be able to provide the same level of accuracy on all types of images.
- The diversity of the dataset used in the testing process is also a significant limitation. If tests are conducted with a limited number of images, it can become difficult to represent all scenarios that may be encountered in the real world. In this case, some special cases or unexpected errors may not be detected during the testing phase. Therefore, testing with large datasets containing as many different types and characteristics of images as possible will increase the reliability of the system.

- The performance of AI models may vary depending on the capture angle of the input images. Model accuracy may decrease, particularly for side-view images or when facial features are not clearly visible due to pose variations or occlusion.

8. Conclusion

This test plan describes the strategies and procedures used to evaluate the functionality, performance, and reliability of the developed system. Throughout the development process, different testing approaches were considered to ensure the system functions as intended. By applying multiple testing methods such as unit testing, integration testing, system testing, performance testing, and user testing, the project aims to verify that each component of the system functions correctly, both on its own and as part of the overall application.

The defined test scenarios provide a structured way to examine the core functions of the system. These functions include image loading, AI-based image processing, preview generation, and downloading the processed image. Systematically testing these features helps developers observe how the system behaves under different conditions and identify potential problems early in the development process.

As a result, potential bugs, performance issues, or unexpected behaviors can be detected and addressed before the system is deployed. Furthermore, the testing process helps evaluate the overall performance and responsiveness of the system while handling different image sizes, processing tasks, and user interactions.

By carefully examining how each module interacts with the others, the development team can ensure smooth communication between components and maintain system stability during operation. Overall, testing plays a crucial role in improving the quality, reliability, and usability of the developed application.

A well-structured test plan not only helps identify technical issues but also contributes to building a more robust and user-friendly system. By following the testing procedures described in this document, the project aims to deliver a reliable and efficient AI-powered image processing application that meets the defined project requirements and provides a consistent user experience.

Requirement Traceability Matrix

Requirement ID	Requirement Description	Related Test Case(s)
FR-01	Image upload	TC-01, TC-02, TC-08, TC-19
FR-02	Object removal	TC-03, TC-09
FR-03	Image enhancement	TC-03, TC-10
FR-04	Style transfer	TC-03, TC-11
FR-05	Facial beautification	TC-03, TC-12
FR-06	Background editing	TC-03, TC-13
FR-07	Image colorization	TC-03, TC-14
FR-08	Result preview	TC-04
FR-09	Image download	TC-05
FR-10	Temporary data handling	TC-07, TC-17
FR-11	Manual mask refinement	TC-15
FR-12	Hybrid editing workflow	TC-16, TC-06, TC-18
FR-13	NSFW Control and restricted control before processing image	TC-17